

Jesse Black

Technical Founder | Staff Software Engineer

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Summary

Building accessible AI communication products and the secure, reliable systems behind them.

Skills

AI Coding Agents: Claude Code, Codex, OpenCode, Antigravity CLI

Agentic Engineering & Quality: harness engineering, agent skills, GitHub Actions CI/CD, quality gates, E2E testing, Dev Containers, Nix flakes

Identity & Security: AuthN/AuthZ, OAuth2/OIDC, SSO, IdP, IAM, RBAC

Developer & Agent Tooling: Rust CLI tools, static analysis, Git diff coverage, documentation tooling

Cloud & Eventing: Azure, Terraform, OpenTofu, Docker, Podman, Event Hub (Kafka-compatible), Service Bus, OpenTelemetry, Azure Monitor Application Insights

AI & Speech Integrations: OpenAI API (GPT models), Azure Speech (speech recognition)

Ecommerce Integrations: App marketplace, OAuth apps, public developer API, Shopify, WooCommerce

Backend: Distributed systems, REST APIs, C#/.NET, Node.js, Python, SQL Server, Redis

Frontend: TypeScript, React, micro frontends (MFE), Tailwind

Experience

Technical Founder, Vibramind LLC – Chicago, IL Oct 2025 – present

Technical founder building an accessibility-first, AI-powered communication app with agentic engineering.

- Founded and built an accessibility-first communication app for Deaf professionals, combining real-time speech-to-text and text-to-speech.
- Built saved responses and GPT-powered smart replies with one-tap playback, context-aware rewrites, and predictive suggestions.
- Built agent harnesses for an agentic engineering workflow, combining custom agent tools, hardened dev sandboxes, and automated quality-gate feedback loops.
- Bootstrapped a secure cloud stack with Terraform, Microsoft Entra External ID, and Azure SQL, optimizing for low burn while designing for scale.

Staff Software Engineer, ShipBob – Chicago, IL Nov 2022 – Oct 2025

Staff engineer building distributed event-driven shipment services and secure iOS micro frontend integration for fulfillment operations at scale.

- Architected and built a fulfillment center shipment service to track and manage shipping orders.
- Modernized shipping order ingestion using Event Hub (Kafka-compatible) and Service Bus to decouple from the monolith and enable tenantization and sharding.
- Enabled all engineering teams to embed micro frontends (MFE) in the iOS app with secure token transfer to enable seamless single sign-on (SSO).
- Strengthened CI/CD quality gates, adding containerized integration testing (Docker) and local database tests to improve end-to-end reliability.
- Mentored two Senior Software Engineers to Staff level.

Senior Software Engineer, ShipBob – Chicago, IL

Dec 2018 – Nov 2022

Senior engineer leading the IAM overhaul and App Marketplace integrations platform, raising the engineering quality bar with automated gates.

- Served as lead engineer for the IAM and App Marketplace teams, driving architecture, execution, and engineer growth through weekly one-on-ones.
- Led the IAM rebuild using IdentityServer for OAuth2 and OpenID Connect (OIDC).
- Enabled fulfillment center managers to create and manage associate logins directly, eliminating IT-ticket-based provisioning as a scaling bottleneck as ShipBob scaled to hundreds of fulfillment centers.
- Led the new App Marketplace with OAuth 2.0, a public Developer API, and iPaaS integration.
- Delivered 6 App Marketplace integrations, including Walmart and Wix.
- Introduced automated code-quality gates (SonarQube) on the App Marketplace team and enforced minimum 80% test coverage across the team's portfolio, achieving the highest overall test coverage in the engineering organization.
- Drove early adoption of C# 8 nullable reference types across the engineering org.
- Implemented and documented many [ShipBob Developer API](#) endpoints, and built the docs system aggregating OpenAPI specs across microservices.

Software Engineer, The Nerdery – Chicago, IL

Nov 2015 – Oct 2018

Digital consultancy engineer delivering custom .NET applications for enterprise clients.

- Built custom web and desktop applications for enterprise clients in a fast-paced consulting environment.
- Worked with client stakeholders to define requirements and deliver .NET/SQL Server solutions.

Projects

docgarden [jesse-black/docgarden](#)

Open-source Rust CLI for repository knowledge systems in agentic engineering repos and GitHub Actions CI.

- Optimized agent token efficiency and made context cheaper by making repository knowledge self-routing.
- Designed BM25F ranked retrieval over repository metadata so agents could find the right plans, skills, and design docs without oversized instruction files.
- Codified repository knowledge standards into deterministic checks for document metadata and token limits on high-traffic files like AGENTS.md.

covgate [jesse-black/covgate](#)

Open-source Rust CLI for Git diff coverage gates in GitHub Actions CI and coding-agent workflows.

- Built a coverage gate that prevents untested code from merging by evaluating only the lines changed in a pull request.
- Supported Rust, JavaScript/TypeScript, and .NET coverage workflows so teams can apply the same quality gate across multiple stacks.
- Added branch, region, and function diff gates alongside line coverage for more precise enforcement on changed code.
- Enabled local diff-gate checks for Codex Cloud and Google Jules task environments, shortening feedback loops for shallow-checkout cloud agent sandboxes.

Education

Rochester Institute of Technology, Bachelor of Science in Computer Science – Rochester, NY

- Minor in Mathematics